12/9/24

Experiment - 6

Aim:
Write program to implement error correction & detection using hamming code make run to input data stream & verify.

Code:

```
def calc_redundant_bits(m):
    for r in range(m+1):
        if(2**r) >= (m+r+1):
            return r

def main():
    data = input("Enter data bits: ").strip()
    m = len(data)
    r = calc_redundant_bits(m)
    n = m+r
    print(f"Number of r needed: {r}")
    print(f"Total frame length: {n}")
    code = ['0'] * (n+1)
    j = 0
    for i in range(1, n+1):
        if(i & (i-1)) != 0:
            code[i] = data[j+1]
            j += 1
    for i in range(r):
        paring_pos = 2**i
        val = 0
        for pos in range(1, n+1):
            if(pos & paring_pos) != 0 and pos !=
                                    parity_pos:
                if code[pos] == '1':
                    val ^= 1
        code[pairing_pos] = str(val)
```

```
transmited_str = " ".join(code[1:][::-1]
print(f"Data to be transmitted : {transmitted_str}")
Change = input("In transmitted bit change").strip().
                                        upper()
if change == "Yes":
    recieved_str = input(f"Enter correpd. data")
    if len(recieved_str) != n:
        print(f" Error: Recieved data be exactly {n}
                                        bits")
        return
    recevied_code = ["0"] + list(recieved_str[::-1])
else:
    received_code = User(code)
error position = 0
for i in range(r):
    paring_pos = 2**i
    val = 0
    for pos in range(1, n+1):
        if (pos & parity_pos) != 0:
            if received_code[pos] == '1':
                val += 1
    if Val != 0
        error_position += pairing_pos
if error_position == 0
    print("In Verification Result : No error detected")
else:
    print("In Verification Result : Errordetected!")
    print(f" Wrong bit at position: {error_position}")
    recieved_code[error_positions] = '0' if
                    recieved_code [error_position] == '1'
    Corrected_str = " ".join(recieved_code [1, n+1][::-1])
                                        else '1'
    print(f" Corrected data stream : {corrected_str}")
if __name__ == "__main__":
    main()
```

Output:

① Enter data bits: 1011100
number of redundant bits (r) needed: 4
Total frame length (m+r): 10
Data to be transmitted: 10111000 11
Transmitted bit change? (Yes/No): Yes
Enter the corrupted data stream (10 nos): 1011101011
Verification Result: Error detected!
Wrong bit is at position: 4
corrected data stream: 1011100011

② Enter data bits: 1011
Number of redundant bits (r) needed: 3
Total data frame length (m+r): 7
Data to be transmitted: 1010101
Transmitted bit change? (YES/NO): NO
Verification Result: No error detected in transmission

Result:

Thus the hamming code was implemented successfully for Error detection & correction.